Zewail City

School of Computational Sciences & Artificial Intelligence

SW402 • SOFTWARE PROJECT MANAGEMENT

Week 6

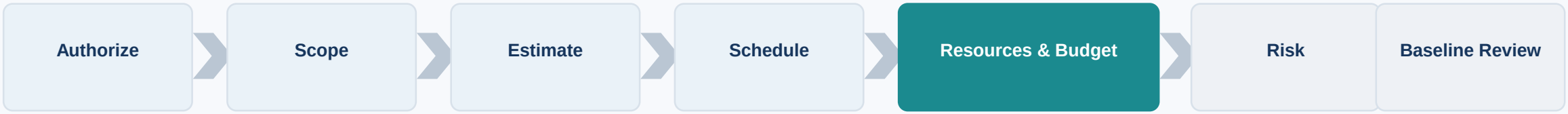
Resource Allocation, Teams, Budget & Optimization

From an ideal schedule to a feasible, staffed and funded plan

Dr. Amr Mansour

Where Week 6 Fits in the Course

The pre-midterm planning story



Week 5 Output

An activity network, critical path, milestones and a schedule that assumes resources are available when needed.

Week 6 Question

Can the planned work actually be staffed, funded and executed without resource conflicts?

Week 6 Output

A resource plan, RACI, resolved over-allocation, time-phased cost view and an optimized feasible schedule.

Learning Outcomes

By the end of this lecture, students should be able to…

1

Identify resource requirements

Translate activities into required roles, skills, quantities and timing.

2

Detect resource conflicts

Recognize loading, over-allocation and bottlenecks.

3

Adjust the schedule

Distinguish smoothing from leveling and explain schedule impact.

4

Assign responsibility

Build a practical RACI and avoid unclear accountability.

5

Build a cost view

Turn resource usage into project cost and a time-phased budget.

6

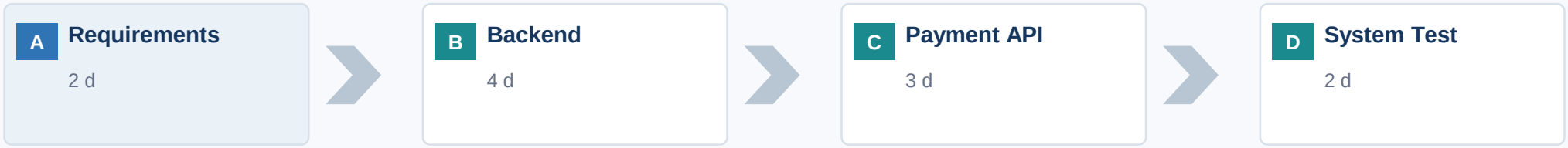
Optimize trade-offs

Evaluate crashing, fast tracking and feasible alternatives.

Warm-up: An “Ideal” Schedule Is Not Yet a Feasible Schedule

2-minute pair task

Week 5 produced this plan:



Question

What information is still missing before you can say this schedule is executable?

- Who performs each activity?
- Are those people available at the same time?
- Do they have the required skills?
- What will the planned resource usage cost?

Today’s Roadmap

Five connected decisions



Hughes/Cotterell alignment: Chapter 8 Resource Allocation + selected people/team concepts from Chapters 11–12.

PART A

Resource Requirements

Start from the activity plan. Identify exactly what each activity needs before assigning names.

What Counts as a Project Resource?

People

BA, developer, tester, UX designer, security specialist

Equipment

Test devices, laptops, servers, lab equipment

Facilities

Meeting room, lab, training room

Software & Services

Cloud environment, CI/CD, payment API, SMS gateway

Consumables / Money

Licenses, travel, cloud usage, external services

Resource Allocation Produces More Than One Schedule

Activity Schedule

Planned start/finish dates for activities

When does the work happen?

Resource Schedule

When and how much of each resource is required

Who/what is needed when?

Cost Schedule

Planned expenditure over time

When will money be spent?

The three views must tell the same story.

Worked Example: ABC Retail — From Activity to Resource Requirement

Activity	Duration	Primary role	Special skill / service
Requirements workshops	3 d	Business Analyst	Facilitation + domain analysis
UI prototype	4 d	UX Designer	Figma / interaction design
Checkout backend	6 d	Backend Developer	Java + transaction logic
Payment integration	4 d	Backend Developer	Gateway API + security review
System testing	5 d	QA Engineer	Integration testing
UAT	2 d	Business Analyst	Sponsor / user availability

Key lesson

Estimate effort first, then state the type, amount and timing of the resources needed to perform it.

Quick Check: Resource or Requirement?

Vote: 3 minutes

A “2 backend developers for 5 days”

B “Complete checkout by 15 Nov”

C “Payment gateway sandbox access”

D “The page must load within 2 seconds”

Classify each statement as: RESOURCE • SCHEDULE CONSTRAINT • QUALITY REQUIREMENT

A Practical Resource Requirement Template

1 **Activity**

What work package / activity?

2 **Role / Resource Type**

What kind of resource?

3 **Skill / Capability**

What competence is required?

4 **Quantity / Effort**

How much?

5 **Timing**

When is it needed?

6 **Availability Constraint**

Any calendar or sharing limit?

Skill Matrix: Availability Is Not the Same as Suitability

Team member	Java	Database	Payment API	Testing	UX
Mona	High	Medium	Low	Medium	Low
Karim	High	High	High	Medium	Low
Salma	Low	Medium	Low	High	Medium
Youssef	Medium	Low	Low	Medium	High

Question 1

Who is the strongest candidate for payment integration?

Question 2

Would you assign the strongest developer to every critical task?

Resource allocation is an optimization problem, not a “best person wins” decision.

Why “2 Developers = Twice the Speed” Is Usually Wrong

Task dependencies

Some work cannot start until another task finishes.

Specialization

Developers may not have the same skills.

Onboarding

New people need project context and setup time.

Integration

Parallel work creates merge and test effort.

Communication

More people create more coordination paths.

Limited workfront

Some activities cannot be split efficiently.

More people ≠ perfectly proportional schedule reduction

Worked Example: Payment Integration Resource Requirement

Activity

Integrate payment gateway

Estimated effort: 160 hours
Planned window: 4 working days
Dependency: checkout backend complete

Resource Requirement

- 1 backend developer (primary)
- 1 security reviewer (8 h)
- Payment provider sandbox
- Test card / test account
- QA support on final day

Availability Check

Developer: available 100%
Security reviewer: only Tue/Thu
Provider sandbox: access confirmed
QA: already booked on another task Wed

The activity estimate did not change — but the feasible schedule may.

Problem: One Security Specialist, Two Activities

Solve in teams: 4 min

Activity	Window	Float	Security effort	Business impact
Payment integration	Days 8–12	0 d	8 h	Blocks checkout release
Admin permissions	Days 8–14	3 d	6 h	Internal admin only

Constraint

The security specialist is available for only 8 hours during Days 8–10.

Which activity gets the specialist first — and why?

Use: criticality • float • downstream impact

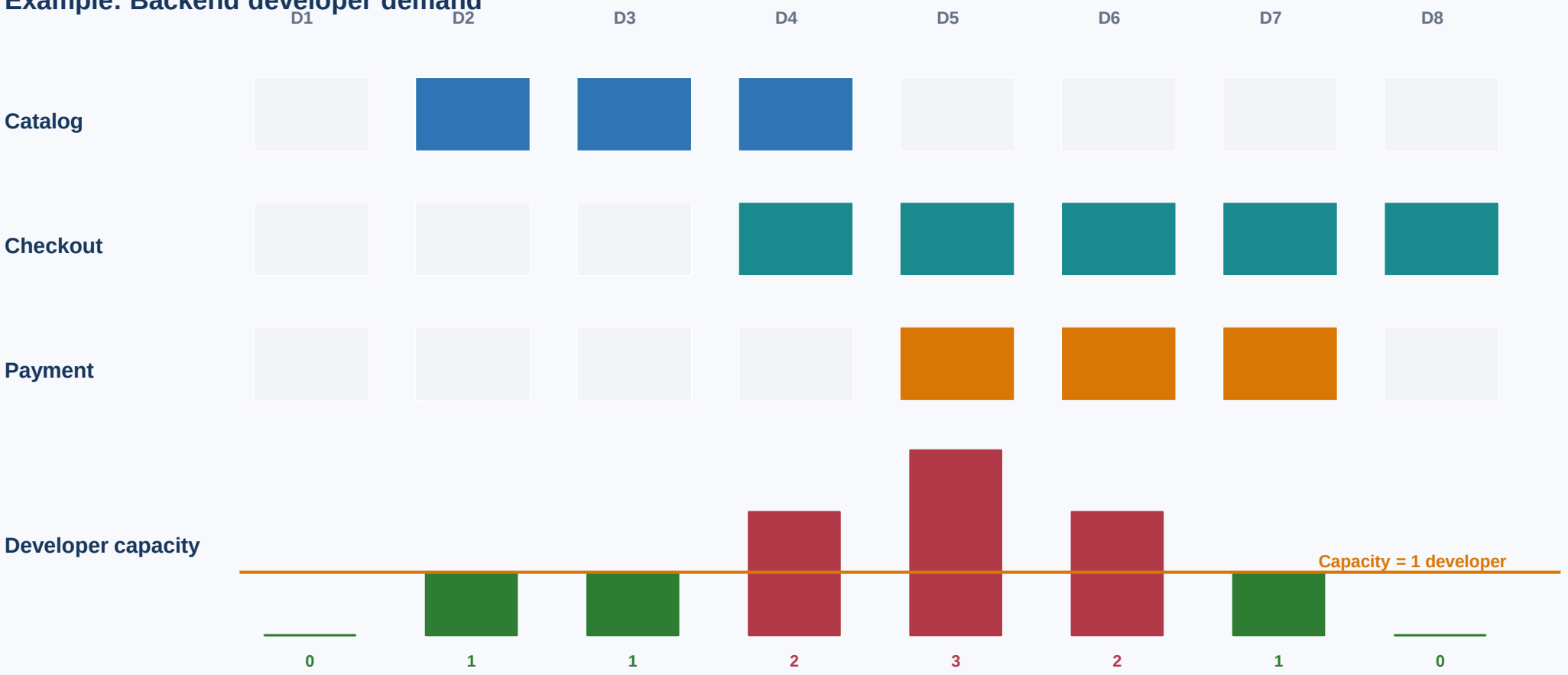
PART B

Resource Loading & Allocation

A schedule becomes feasible only when resource demand fits actual capacity.

Resource Loading: How Much Demand Exists at Each Point in Time?

Example: Backend developer demand



Interpretation

Demand above capacity = over-allocation. The plan may be logically valid but operationally impossible.

Over-allocation: The Same Resource Is Booked Twice



Conflict

Tue–Thu require the same developer on two activities at the same time.

Move work

Use available float

Reassign

Use another qualified resource

Split / sequence

Change the execution plan

Add capacity

Overtime / contractor / extra resource

Problem: What Would You Move?

Individual: 3 min

Activity	Duration	Resource	Float	Planned start
A: API integration	3 d	Dev-1	0 d	Day 6
B: Admin reports	2 d	Dev-1	4 d	Day 6
C: UAT scripts	2 d	BA	2 d	Day 7

Resource Constraint

Dev-1 can work on only one activity at a time.

Choose the first corrective action and justify it.

Priority clue: protect zero-float work first

Solution: Use Float Before Changing the Project Finish Date

Before

A: API integration Day 6–8 • Float 0
B: Admin reports Day 6–7 • Float 4

Result: over-allocation on Days 6–7



After

A: API integration Day 6–8
B: Admin reports Day 9–10

Result: conflict removed; project finish unchanged

This is resource smoothing

When Resources Compete: Useful Priority Rules

1	**Critical activities first**	Zero float means delay can move the project finish.
2	**Least float next**	Protect near-critical paths before flexible work.
3	**Downstream impact**	Prefer work that unlocks many successors.
4	**Specialist scarcity**	Use scarce expertise where substitution is hardest.
5	**Business / risk impact**	Consider value, contractual dates and risk exposure.

Caution

Priority rules guide judgment; they do not replace dependency logic, skill requirements, or management decisions.

Resource Smoothing vs Resource Leveling

Resource Smoothing

Goal: reduce peaks / improve utilization

How: move non-critical activities within available float

Project finish date: normally unchanged

Best when: time is the stronger constraint

Resource Leveling

Goal: make demand fit limited resource capacity

How: delay or resequence work when capacity is insufficient

Project finish date: may change

Best when: resource availability is the stronger constraint

Smoothing uses float • Leveling may create a new schedule

Is the Project Time-Constrained or Resource-Constrained?

Regulatory launch on 1 January	**Time-constrained**	Deadline dominates; adjust resources/strategy to protect date.
Only one certified security engineer exists	**Resource-constrained**	Capacity dominates; schedule may extend.
Marketing campaign must launch before event	**Time-constrained**	Use smoothing, reassignment, or added capacity.
Shared UX specialist available 2 days/week	**Resource-constrained**	Level work around the calendar.

Worked Example: Leveling a Shared Developer

Activity	Pred.	Dur.	Resource
A Requirements	—	2 d	BA
B Backend	A	4 d	Dev-1
C Payment API	A	3 d	Dev-1
D System Test	B, C	2 d	QA

Ideal network schedule

A (2 d) → B and C in parallel → D. Project duration = 8 days.

But…

B and C both require Dev-1. One person cannot execute 7 days of work in the same 4-day window.

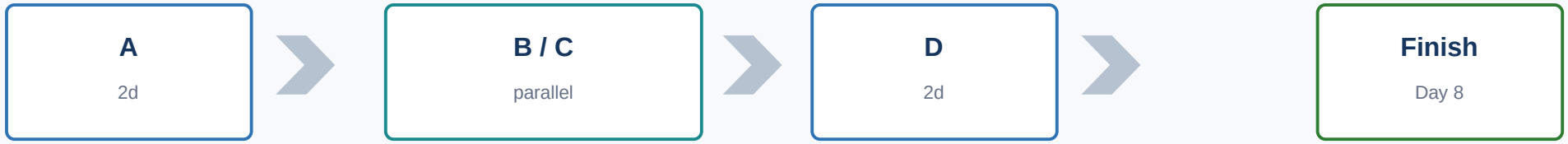


Leveled schedule

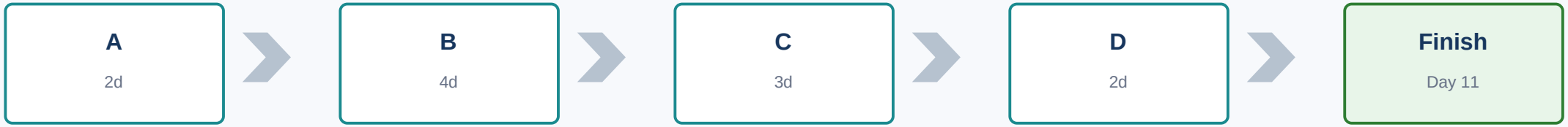
A → B → C → D (or A → C → B → D). Project duration = 11 days.

Resource Constraints Can Create a New Critical Sequence

Dependency logic says:



Resource reality says:



Resource feasibility can change which sequence controls completion

Student Problem: Level the Smart Clinic Schedule

Pairs: 5 min

Activity	Pred.	Dur.	Resource
A Requirements	—	2 d	BA
B Booking module	A	4 d	Dev-A
C Queue module	A	3 d	Dev-A
D Notifications	A	2 d	Dev-B
E Integration test	B, C, D	2 d	QA

Task 1

Find the ideal duration if resource limits are ignored.

Task 2

Dev-A cannot work on B and C simultaneously. Produce a feasible sequence and new duration.

Do not move D unless necessary

Solution: Smart Clinic Resource Leveling

Ideal Plan

A (2d)
Then B (4d), C (3d), D (2d) in parallel
Then E (2d)

Duration = 2 + 4 + 2 = 8 days

Feasible Plan

A (2d)
Then B (4d) → C (3d) for Dev-A; D (2d) can run in parallel
Then E (2d)

Duration = 2 + 7 + 2 = 11 days

What changed?

- No scope changed.
- No activity effort changed.
- Only resource availability changed — but the project duration increased by 3 days.

Management decision

Accept the 11-day plan, add qualified capacity, or redesign/overlap the work — each option has cost/risk consequences.

Resource Calendars: “Available” Is a Calendar, Not a Yes/No Field

Resource	Working calendar	Capacity	Constraint
Backend Dev	Mon–Fri	100%	No leave
Security Specialist	Tue & Thu	25%	Shared across projects
Sponsor	Mon 2–4 PM	—	UAT approvals only
Test Device Lab	Wed–Fri	1 unit	Booked by another team Mon–Tue

Why it matters

A 2-day task may take a full week if the required specialist is available only one day at a time.

Duration depends on effort + capacity + calendar + dependencies

Microsoft Project Workflow: From WBS to a Feasible Resource Schedule

1 Enter / confirm tasks

Use Week 5 WBS and dependencies

2 Create resources

Names, max units, calendars, rates

3 Assign resources

Match skills to activities

4 Inspect over-allocation

Resource Sheet / Resource Usage / Team Planner

5 Resolve conflicts

Smoothing, leveling, reassignment, scope/schedule decision

6 Review finish + cost

Do not accept automatic leveling blindly

PART C

Teams & Responsibility

Allocation answers “who is available?” Responsibility answers “who owns the result?”

RACI: Make Responsibility Visible



Responsible

Does the work



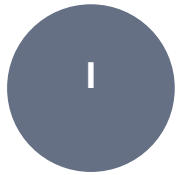
Accountable

Owns the result / decision



Consulted

Provides input before action



Informed

Kept updated

Rule of thumb

Each deliverable should have clear accountability. Multiple “A” assignments usually signal unclear ownership.

Worked RACI: ABC Retail

Deliverable	PM	BA	Dev Lead	QA	Sponsor
Requirements baseline	A	R	C	I	C
Checkout module	A	C	R	C	I
Payment integration	A	C	R	C	I
System test report	A	C	C	R	I
UAT acceptance	C	R	I	C	A

Observe

The PM is not automatically Responsible for every deliverable.

Observe

Sponsor accountability appears where business acceptance is required.

RACI Problem: Fix the Responsibility Matrix

Pairs: 4 min

Deliverable	PM	BA	Dev	QA	Sponsor
Prototype approval	A	R	C	I	A
System testing	A	C	R	R	I
Final acceptance	R	C	I	C	I

Find the issues

- Too many A?
- Missing A?
- Too many R?
- Wrong business owner?

Propose a corrected matrix — there may be more than one reasonable answer.

Team Selection: Balance Capability, Availability and Collaboration



Best assignment ≠ strongest technical person in isolation

Team Growth Increases Communication Paths

Communication channels = n(n − 1) / 2

3 people	5 people	8 people	10 people
3 channels	10 channels	28 channels	45 channels

Management implication

Adding people increases potential capacity, but also coordination overhead. Organize work, interfaces and communication deliberately.

Software Teams Need Different Leadership at Different Moments

Unclear / new team	More direction	Clarify goals, roles, standards
Developing capability	Coaching	Give feedback and remove blockers
Experienced team	Delegation	Trust ownership and focus on outcomes
Conflict / pressure	Active facilitation	Resolve priorities and protect collaboration

PART D

Cost & Budget

Once resources are assigned, the schedule can be translated into planned expenditure over time.

From Resource Usage to Project Cost

Labor

Role rates × planned effort
Developers, BA, QA, PM

Non-labor

Tools, cloud, services
Licenses, SMS, payment fees

Indirect / Shared

Allocated overhead
Shared facilities, administration

Contingency

Known uncertainty allowance
Planned response to uncertainty

Worked Cost Example: ABC Retail — Same Week 4 Total, Better Detail

Resource / cost item	Planned amount	Rate	Cost (EGP)
Project Manager	2.0 PM	40,000 / PM	80,000
Business Analyst	1.5 PM	35,000 / PM	52,500
Developers	4.5 PM	42,000 / PM	189,000
QA Engineer	0.75 PM	30,000 / PM	22,500
UX support	0.20 PM	30,000 / PM	6,000
Labor total			350,000
Non-labor			70,000
Contingency 12%			50,400
Estimated total			470,400

Continuity

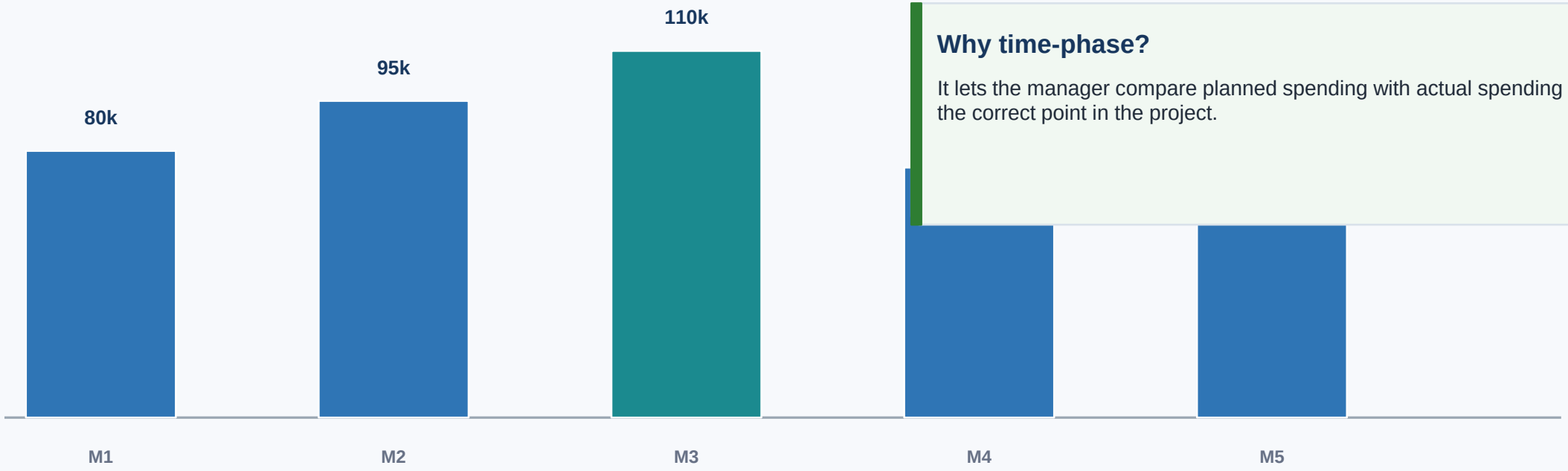
Week 4 estimated 470,400 EGP. Week 6 explains where that number comes from and when it will be spent.

Important

A resource change can affect both the schedule and the cost baseline.

Time-Phased Budget: Cost Must Be Connected to Time

Planned cost before contingency = 420k EGP



Why time-phase?

It lets the manager compare planned spending with actual spending at the correct point in the project.

Cost, Budget and Price Are Not the Same

Cost

What the project is expected to consume

Example:
Labor + services + infrastructure

Budget

Approved funding available to manage the project

Example:
Cost baseline + approved reserves/policy

Price

What a supplier charges / customer pays

Example:
May include profit, tax, commercial terms

Do not report “the project costs 600k” when you actually mean the customer price is 600k.

Problem: Build a Mini Cost Schedule

Solve: 4 min

Activity	Month	Resource cost
Requirements	M1	30,000 EGP
Development	M1–M2	120,000 EGP total; 50/50
Testing	M2	35,000 EGP
Deployment	M3	25,000 EGP

Your task

Calculate planned project cost in M1, M2 and M3, then the cumulative cost at the end of each month.

M1 = ? • M2 = ? • M3 = ?

Do Not Hide Uncertainty Inside Every Task Estimate

Poor practice

Every team member secretly adds “extra days” to every activity.

Result: hidden padding, unclear ownership, hard-to-explain baseline.

Better practice

Use transparent assumptions and an explicit contingency/reserve policy.

Result: traceable estimates and better management decisions.

Week 7 will connect contingency to specific risks

PART E

Optimization & Trade-offs

When the feasible plan misses the target, management must choose what to change.

Crashing: Spend More to Reduce Duration on Selected Activities

Concept

Add capacity or use a more expensive option to shorten activity duration.

Examples:
- Add a qualified contractor
- Pay overtime
- Use a faster external service
- Buy automation / tooling

Rules

Crash only where it can change the project finish.

Check:
- Critical path impact
- Extra cost per day saved
- Quality / coordination risk
- New critical path after compression

Crash cost slope = (Crash cost − Normal cost) / (Normal duration − Crash duration)

Worked Example: Which Activity Should You Crash First?

Activity	Path?	Normal	Crash	Extra cost	Days saved	Cost/day saved
A Requirements	Critical	4 d	3 d	6,000	1	6,000
B Backend	Critical	6 d	4 d	18,000	2	9,000
C UI	Non-critical	5 d	3 d	5,000	2	2,500

Temptation

Crash C because it is cheapest per day.

Correct logic

If C is not controlling project completion, crashing it may save zero project days.

Decision

Start with a crashable critical activity; compare A vs B and recalculate after each change.

Fast Tracking: Overlap Work That Was Previously Sequential

Before

Design
4d

Build
6d

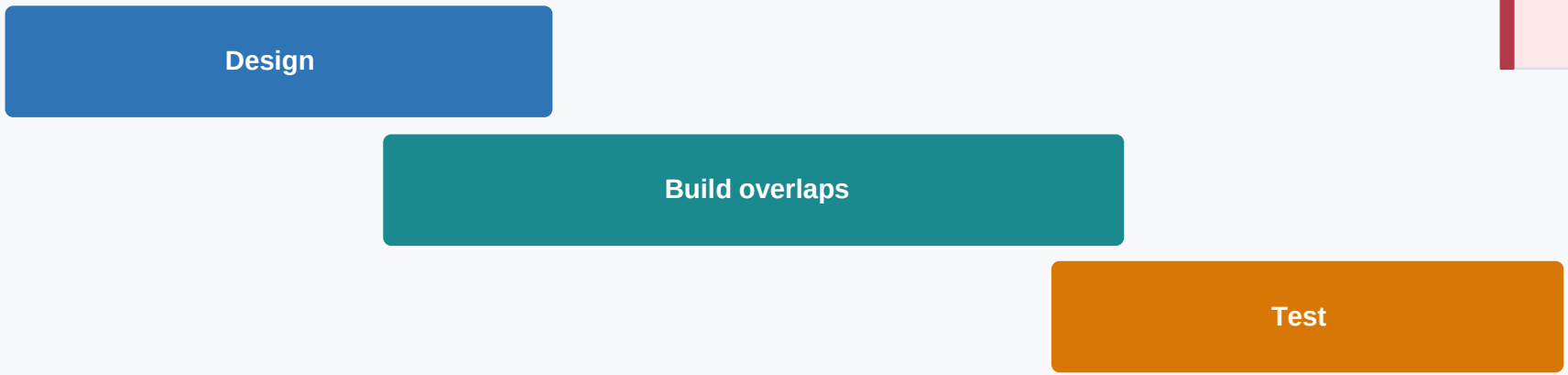
Test
4d

Trade-off

Potentially shorter schedule

But higher risk of rework, coordination problems and change propagation.

After: start Build before all Design is complete



Management Challenge: Deadline Is 2 Weeks Earlier

Decision board: 6 min

Option A — Add contractor

Cost +45,000 EGP
Expected saving: 8 days
Risk: onboarding / integration

Option B — Fast track

Cost +5,000 EGP
Expected saving: 6 days
Risk: high rework probability

Option C — Reduce scope

Move analytics dashboard to Release 2
Expected saving: 10 days
Risk: sponsor dissatisfaction

Which option would you recommend? What information is missing before approval?

Integrated Case: Smart Clinic — From Schedule to Resource Decision

Fact	Value
Current plan	11 days after leveling
Target	8 days
Bottleneck	One developer performs Booking + Queue modules
Budget headroom	40,000 EGP
Alternative	Contract developer: 30,000 EGP for 4 days
Risk	Contractor needs 1 day onboarding

Decision 1

Would the contractor actually restore the 8-day target? Show the logic.

Decision 2

What new resource / integration risks appear?

Decision 3

What would you record in the resource plan and cost baseline?

Week 6 Summary: The Feasible Plan



Core principle

A good project plan is not the earliest possible schedule. It is a schedule that is logically valid, resource-feasible, financially supported, clearly owned and acceptable to stakeholders.

Week 6 Practical Milestone — M6 Resource & Cost Baseline

Resource Plan

Roles, skills, quantities, calendars and availability

RACI Matrix

Clear R/A/C/I for major deliverables

Resource-loaded Schedule

Assignments entered in Microsoft Project

Conflict Analysis

Evidence of over-allocation + how it was resolved

Cost / Budget

Resource costs + non-labor costs + time-phased view

Optimized Schedule

Explain any leveling, reassignment, crashing or fast tracking

Use the same semester project — do not create a new case

Exit Quiz — 5 Questions

1. **Moving a non-critical task within float to reduce a resource peak is called…**

2. **If resource capacity forces the project end date to move, which technique is occurring?**

3. **In RACI, who owns the final result / decision?**

4. **Why can adding a developer fail to halve the duration?**

5. **What is the key difference between crashing and fast tracking?**

Answers: 1) smoothing 2) leveling 3) Accountable 4) dependencies/coordination/specialization 5) add cost/capacity vs overlap work/risk rework

References & Course Alignment

Primary textbook alignment

Hughes, Cotterell & Mall, Software Project Management — Resource Allocation chapter: nature of resources; identifying requirements; scheduling resources; critical paths; costs; resource schedule; cost schedule; scheduling sequence.

Selected people/team alignment

Managing people and working in teams: staff selection, motivation, group working, leadership and team organization — used only where they support the Week 6 resource/team milestone.

Teaching supplements

Resource smoothing vs leveling, RACI, schedule compression and Microsoft Project workflow are included as practical PM tools to support the official course outcomes and semester project.